

Rohan Siva

rohansiva@utexas.edu • (669) 294-7436 • [linkedin.com/in/rohan-siva](https://www.linkedin.com/in/rohan-siva) • github.com/Rohan-Siva • [Google Scholar](https://scholar.google.com/citations?user=rohansiva)

EDUCATION

University of Texas at Austin

B.S. Electrical and Computer Engineering Honors

Expected: May 2027

GPA: 3.98/4.0

Relevant Coursework: Data Structures, Algorithms, Machine Learning, Data Science Principles, Data Science Lab, Probability and Statistics, Object-Oriented Design, Discrete Math, Multivariable Calculus, Linear Algebra, Differential Equations, Embedded Systems

SELECTED PUBLICATIONS

Siva, R., Bhatt, N. P., Yang, Y., Lee, S., Gadde, N., Ellis, C., Velasquez, A., Wang, Z., & Topcu, U. "What Objects Enable, Not What They Are: Functional Latent Spaces for Affordance Reasoning," In Proceedings, **Conference on Robot Learning (CoRL), 2026.**

Siva, R., Cheung, K., Li, L., & Sundaram, G. "kRAIG: A Natural Language-Driven Agent for Automated DataOps Pipeline Generation," In Proceedings, **Conference on Neural Information Processing Systems (NeurIPS), 2026.**

Bhatt, N. P., Yang, Y., Siva, R., Samineni, P., Milan, D., Wang, Z., & Topcu, U. "VLN-Zero: Rapid Exploration and Cache-Enabled Neurosymbolic Vision-Language Planning for Zero-Shot Transfer in Robot Navigation," In Proceedings, **NeuS, 2026. (Spotlight)**

Bhatt, N. P., Yang, Y., Siva, R., Milan, D., Topcu, U., & Wang, Z. "Know Where You're Uncertain When Planning with Multimodal Foundation Models: A Formal Framework," In Proceedings, **Conference on Machine Learning and Systems (MLSys), 2025. (Oral)**

Bhatt, N. P., Li, P., Gupta, K., Siva, R., Milan, D., Hogue, A. T., Chinchali, S. P., Fridovich-Keil, D., Wang, Z., & Topcu, U. "UNCAP: Uncertainty-Guided Planning Using Natural Language Communication for Cooperative Autonomous Vehicles," In Proceedings, **International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2026. (Oral + Best Paper Finalist)**

EXPERIENCE

Decagon | Research Intern | San Francisco, CA

June 2026 - Present

- Built native voice inference engine with packed prefill, continuous batching, and paged KV caching; achieved 9.6× throughput
- Architected batched LM and stateful AudioVAE decode with CUDA graphs and local GPU state, cutting audio E2E latency 9.1×
- Designed novel RL algo for flow-matching speech models to optimize voice naturalness and control tags, with 98.2% stability
- Shipped audio signal-to-noise ratio gate, filtering ~10% of noisy audio segments, and improving turn detection by 15%

UT Austin Center for Autonomy | Researcher | Austin, TX

Aug 2024 – Present

- Co-authored 8 papers on autonomous planning, perception, and computer vision: 2 oral presentations and Best Paper Nomination
- Developed latent-space failure prediction and steering for $\pi 0.5$ VLAs, evaluated on 1,151 LIBERO rollouts with 91.5% success
- Built function-based robot learning methods achieving 25–36% gains over VLM baselines, and 100–350× faster inference

Mercor | Machine Learning Engineer Intern | Remote

Aug 2025 - Dec 2025

- Built a RAG-based evaluation pipeline processing 50+ OpenAI MLE tasks daily, reducing manual review time by 85%
- Overhauled LLM evaluation to achieve sub-200 ms latency and cut compute costs by 30% through optimized embedding caching

Statistical Learning & AI Group Lab @ UT Austin | Researcher

May 2025 – Aug 2025

- Optimized speculative decoding for GRPO, increasing acceptance rate by 15% via training trajectory-conditioned EAGLE drafter
- Explored custom GRPO and PPO implementations for NanoChat and improved post-RL GSM8K performance by ~12%

Cisco Systems | Machine Learning Engineer Intern | San Francisco, CA

May 2025 – Aug 2025

- Created an agent to convert natural language into ETL pipelines with direct enterprise PostgreSQL data loading, adopted by 9+ internal teams, decreasing pipeline-creation tickets by 80% per quarter and reducing setup time by 70%. Published as a paper.
- Built full-stack microservice with React, ElasticSearch, Bedrock, and AWS Lambda for deployment, delivering 99.7% uptime

PROJECTS

Vision-Language Action Safety Prediction Module

Sep 2026

- Built VLA latent-space probes to predict occlusion-induced failures across 4,000 matched normal/occluded LIBERO rollouts
- Fine-tuned GR00T N1.7 on LIBERO and real-world robot data with DeepSpeed ZeRO-2, cutting model-state memory by ~63%

MoodScribe: Emotion-based Chatbot | [Github](#)

Nov 2024

- Architected a full-stack web app with CyberPunk-themed UI (HTML/CSS) to generate sentiment-based Instagram reply
- Fine-tuned Llama 3.1 using Unsloth and quantized LoRA adapters for sentiment analysis, cutting inference latency by 44%

SKILLS & INTERESTS

Languages: Python, Java, C/C++, JavaScript, SQL, TypeScript, HTML/CSS, Swift

Frameworks & Tools: React.js, Node.js, Flask, Docker, Kubernetes, PostgreSQL, AWS, Git, CI/CD, Azure, GCP, NumPy, Pandas

AI/ML: PyTorch, Transformers, OpenCV, World Models, GRPO/PPO, Sklearn, LangChain, Transformers, Carla, Habitat